Huawei Access Node BAS-ACC-28 Optical Transport Failure Resolution SOP

Node: BAS-ACC-28 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2344

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps to address Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on Huawei Access Node BAS-ACC-28 at Basavanagudi POP. The failure domain is specific to the Access Role and Vendor Equipment, where the Optical Transport Network (OTN) is responsible for transmitting optical signals between nodes. The OTN employs a protocol stack consisting of Optical Layer (Layer 0), Optical Multiplex Section (OMS), Optical Transport Section (OTS), and Optical Channel (OCh). In case of SFP/XFP degradation, a reduction in Rx/Tx power loss can lead to signal degradation or even loss, resulting in critical alarm ALARM_2344. The underlying protocol mechanics involve the exchange of Optical Supervisory Channel (OSC) signals between adjacent nodes, which monitor the health of the optical link. The Huawei Access Node employs the OTN protocol stack, and failure to maintain the optical link can cause signal degradation, leading to the critical alarm.

Pre-requisite Checks

- Check: Verify the presence of redundant links and ensure the secondary link is available., Status: Dispatch remote hands to the Basavanagudi POP for assistance, if necessary.

- Check: Ensure adequate power supply to the node and check for any nearby infrastructure issues., Status: Verify that all necessary safety protocols are followed during the troubleshooting process.

- Check: Review the node's alarm history and confirm the occurrence of the critical ALARM_2344., Status: Document the incident and update the maintenance records.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts or damage	Medium
Rx/Tx power loss due to misconfiguration or equipment failure	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-link, Expected Output: Optical link status and Rx/Tx power levels, Description: This command displays the current state of the optical link, including the Rx/Tx power levels, which can help identify potential issues with the SFP/XFP.

Command: display optical-oms, Expected Output: Optical multiplex section status, Description: This command displays the status of the
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Optical Multiplex Section (OMS), which can help identify issues with
the optical link.

Command: display otn-osc, Expected Output: Optical supervisory channel
status, Description: This command displays the status of the Optical
Supervisory Channel (OSC), which monitors the health of the optical
link.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP, Command: replace optical-module , Description: This step involves replacing the faulty SFP/XFP with a new one. The exact command may vary depending on the slot and module name.

Step 2: Step: Verify and adjust the Rx/Tx power levels, Command: set optical-power , Description: This step involves verifying and adjusting the Rx/Tx power levels to ensure they are within the recommended range.

Step 3: Step: Check and repair fiber cuts or damage, Command: display optical-link , Description: This step involves checking and repairing fiber cuts or damage, if any.

Post-Check Verification

```
Command: display optical-link, Expected Output: Optical link status
and Rx/Tx power levels, Description: This command verifies that the
optical link is stable and the Rx/Tx power levels are within the
recommended range.

Command: display otn-osc, Expected Output: Optical supervisory channel
status, Description: This command verifies that the Optical
Supervisory Channel (OSC) is functioning correctly.
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Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: replace optical-module , Description: This step involves restoring the original SFP/XFP.

- Step: Revert the Rx/Tx power level adjustments, Command: set optical-power , Description: This step involves reverting the Rx/Tx power level adjustments.